Day 14 Study Notes

Thu 26 Nov 2026 · Sprint 60 · JEE Main 2027

Part	Time	What to do
Study	3 h 30 min	Physics: Ray optics (part 3: prism and instruments) (75 min) · Maths: Quadratic equations (75 min) · Chemistry: Hydrocarbons (part 2: alkynes and benzene) (60 min)
Exam	1 h 15 min	The Day 14 exam in the Sprint 60 app: 30 questions, timed, no notes.
Review	1 h 15 min	Open every wrong and skipped question in the app, read its solution, tag the reason, then use Retry my mistakes.

How to use each chapter note. Read it once without stopping. Cover the formulas and write them from memory. Work the solved examples on paper before reading the working. Then read the NCERT section listed at the end.

Physics · Ray optics (part 3: prism and instruments)

This finishes ray optics: the prism, dispersion, scattering, and the microscope and telescope. Instrument questions are formula-based and quick marks.

1. Prism

- For a prism of angle A: $r_1 + r_2 = A$ and the deviation is $\delta = i + e - A$.
- As i increases, δ first falls, reaches a minimum δ_m, then rises.
- At **minimum deviation**: $i = e$, $r_1 = r_2 = \dfrac{A}{2}$, and the ray inside the prism is parallel to the base.
- $n = \dfrac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\frac{A}{2}}$.
- **Thin prism** (small A): $\delta = (n-1)A$.
- No emergent ray (total internal reflection at the second face) if $A > 2C$.

2. Dispersion

- n is larger for shorter wavelengths, so violet deviates most and red least.
- Angular dispersion $= \delta_v - \delta_r = (n_v - n_r)A$ for a thin prism.
- Dispersive power $\omega = \dfrac{n_v - n_r}{n_y - 1}$, where n_y is for yellow (mean) light. It depends only on the material.
- Two thin prisms give **dispersion without deviation** if $(n-1)A + (n'-1)A' = 0$, and **deviation without dispersion** if $(n_v - n_r)A + (n'_v - n'_r)A' = 0$.

3. Scattering

- Rayleigh scattering by particles much smaller than the wavelength varies as $\dfrac{1}{\lambda^4}$, so blue is scattered far more than red.
- The sky is blue; the sun looks red at sunrise and sunset because blue has been scattered out over the long path. Clouds look white because their droplets are large and scatter all colours alike.

4. Simple microscope (magnifying glass)

- Image at the near point $D = 25$ cm: $m = 1 + \dfrac{D}{f}$. Image at infinity (relaxed eye): $m = \dfrac{D}{f}$.

5. Compound microscope

- Objective: short focal length, small aperture. Eyepiece: larger focal length. Both convex.
- Final image at infinity: $m \approx \dfrac{L}{f_o} \cdot \dfrac{D}{f_e}$, where L is the tube length.
- Final image at the near point: $m = \dfrac{v_o}{u_o}\left(1 + \dfrac{D}{f_e}\right)$ in magnitude.

6. Astronomical telescope

- Objective: long focal length, large aperture. Eyepiece: short focal length.

- Normal adjustment (final image at infinity): $m = \dfrac{f_o}{f_e}$, and the tube length is $f_o + f_e$.
- Final image at the near point: $m = \dfrac{f_o}{f_e}\left(1 + \dfrac{f_e}{D}\right)$, tube length $f_o + u_e$.
- A larger objective aperture gathers more light and resolves finer detail.

Solved example 1. A prism of angle 60° gives a minimum deviation of 30°. Find n.

$n = \dfrac{\sin 45^\circ}{\sin 30^\circ} = \dfrac{1/\sqrt{2}}{1/2} = \sqrt{2}$.

Solved example 2. A telescope has $f_o = 100$ cm and $f_e = 5$ cm. Find its magnifying power and length in normal adjustment.

$m = \dfrac{100}{5} = 20$. Length $= 105$ cm.

Traps. Violet deviates more than red, not less. Use $\delta = (n-1)A$ only for small angles. In a telescope the objective has the longer focal length; in a microscope it has the shorter. $1 + \frac{D}{f}$ is for the image at 25 cm, $\frac{D}{f}$ for the image at infinity.

Mathematics · Quadratic equations

Quadratic equations give one question in almost every paper, usually on sums and products of roots, the nature of roots or the location of roots. This chapter was originally planned for Day 21 and moved here.

1. Roots and the discriminant

- $ax^2 + bx + c = 0$ ($a \neq 0$) has roots $x = \dfrac{-b \pm \sqrt{D}}{2a}$, with $D = b^2 - 4ac$.
- $D > 0$: two real, distinct roots. $D = 0$: equal roots. $D < 0$: two complex conjugate roots.
- With rational coefficients, a perfect-square D gives rational roots; otherwise irrational roots come in conjugate pairs $p \pm \sqrt{q}$.
- With real coefficients, complex roots always come in conjugate pairs, so a quadratic cannot have exactly one complex root.
- If $a + b + c = 0$, then $x = 1$ is a root.

2. Sum and product of roots

- $\alpha + \beta = -\dfrac{b}{a}$ and $\alpha\beta = \dfrac{c}{a}$.
- $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$.
- $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$.
- $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$.
- $\dfrac{1}{\alpha} + \dfrac{1}{\beta} = \dfrac{\alpha + \beta}{\alpha\beta}$.
- The equation with roots α, β: $x^2 - (\alpha + \beta)x + \alpha\beta = 0$.
- Equation with roots $\dfrac{1}{\alpha}$, $\dfrac{1}{\beta}$: swap a and c. Roots $k\alpha$, $k\beta$: replace x by $\dfrac{x}{k}$. Roots $\alpha + k$, $\beta + k$: replace x by $x - k$.

3. Common root

- If $a_1x^2 + b_1x + c_1 = 0$ and $a_2x^2 + b_2x + c_2 = 0$ share one root, subtract a multiple of one from the other to remove x^2, find the root, and substitute.
- Both roots common: $\dfrac{a_1}{a_2} = \dfrac{b_1}{b_2} = \dfrac{c_1}{c_2}$.

4. Graph, maximum and minimum

- $y = ax^2 + bx + c$ is a parabola opening up if $a > 0$, down if $a < 0$, with vertex at $x = -\dfrac{b}{2a}$, $y = -\dfrac{D}{4a}$.
- For $a > 0$ the minimum value is $-\dfrac{D}{4a}$; for $a < 0$ that is the maximum.
- $ax^2 + bx + c > 0$ for all real x exactly when $a > 0$ and $D < 0$.

5. Location of roots

For $f(x) = ax^2 + bx + c$:

- Both roots positive: $D \geq 0$, $\alpha + \beta > 0$, $\alpha\beta > 0$. Both negative: $D \geq 0$, $\alpha + \beta < 0$, $\alpha\beta > 0$.
- Roots of opposite signs: $\alpha\beta < 0$, that is $\dfrac{c}{a} < 0$.
- A number k lies between the roots: $a\, f(k) < 0$.
- Both roots greater than k: $D \geq 0$, $a\, f(k) > 0$ and $-\dfrac{b}{2a} > k$.

6. Equations with $|x|$

- Put $t = |x| \geq 0$, solve for t, keep only $t \geq 0$, then each positive t gives two values of x.

> **Solved example 1.** If α, β are the roots of $x^2 - 4x + 2 = 0$, find $\alpha^3 + \beta^3$.
> $\alpha + \beta = 4$, $\alpha\beta = 2$. $\alpha^3 + \beta^3 = 64 - 3 \times 2 \times 4 = 40$.

> **Solved example 2.** For which $a \neq 1$ do $x^2 + ax + 1 = 0$ and $x^2 + x + a = 0$ have a common root?
> Subtracting: $(a-1)x - (a-1) = 0$, so $x = 1$. Then $1 + a + 1 = 0$, $a = -2$.

Traps. $\alpha + \beta = -\dfrac{b}{a}$: watch the minus sign. "Real roots" allows $D = 0$; "distinct real roots" needs $D > 0$. When $t = |x|$, reject negative t. A quadratic needs $a \neq 0$: check the case where the leading coefficient contains the parameter.

Chemistry · Hydrocarbons (part 2: alkynes and benzene)

Part 1 covered alkanes and alkenes. Today covers alkynes and benzene, especially electrophilic substitution and directing groups, which are asked almost every year.

1. Alkynes: preparation

- Calcium carbide and water: $CaC_2 + 2H_2O \rightarrow C_2H_2 + Ca(OH)_2$.
- Double dehydrohalogenation of a vicinal dihalide with alcoholic KOH, then $NaNH_2$.

2. Alkynes: reactions

- **Acidity of terminal alkynes:** the H on an sp carbon is weakly acidic. Terminal alkynes react with Na or $NaNH_2$, give a white precipitate with ammoniacal $AgNO_3$ (Tollens') and a red one with ammoniacal Cu_2Cl_2. Internal alkynes such as but-2-yne do not. This distinguishes but-1-yne from but-2-yne.
- **Hydrogenation:** H_2/Ni goes all the way to the alkane; Lindlar's catalyst stops at the cis alkene; Na/liquid NH_3 gives the trans alkene.
- **Addition of HX** happens twice, following Markovnikov's rule, giving a geminal dihalide.
- **Hydration** with dilute H_2SO_4 and $HgSO_4$ gives an enol that turns into a carbonyl compound. Ethyne gives ethanal; every other alkyne gives a ketone (propyne gives propanone).
- **Cyclic polymerisation:** ethyne passed through a red-hot iron tube at 873 K gives benzene.

3. Benzene

- Planar, all six carbons sp^2, all C–C bonds equal (139 pm, between single and double). 12 σ bonds and a delocalised ring of 6 π electrons.
- Very stable because it is aromatic, so it prefers substitution, which keeps the ring, to addition, which destroys it.
- Addition happens only under harsh conditions: H_2/Ni at high temperature gives cyclohexane; Cl_2 in UV light gives benzene hexachloride.

4. Electrophilic aromatic substitution

Three steps: the electrophile is generated; it attacks the ring to form a carbocation (the arenium ion or σ complex, the slow step); the ring loses H^+ to regain aromaticity.

Reaction	Reagents	Electrophile	Product from benzene
Nitration	conc. HNO_3 + conc. H_2SO_4	NO_2^+	nitrobenzene
Halogenation	Cl_2 + $FeCl_3$ (or $AlCl_3$)	Cl^+	chlorobenzene
Sulphonation	fuming H_2SO_4	SO_3	benzenesulphonic acid
Friedel–Crafts alkylation	RCl + anhydrous $AlCl_3$	R^+	alkylbenzene (CH_3Cl gives toluene)
Friedel–Crafts acylation	RCOCl + anhydrous $AlCl_3$	RCO^+	aryl ketone (acetophenone)

5. Directing groups

Group on the ring	Effect on rate	Directs to

–OH, $–NH_2$, $–OCH_3$, $–NHCOCH_3$, alkyl	activating	ortho and para
–F, –Cl, –Br, –I	deactivating	ortho and para
$–NO_2$, –CN, –CHO, –COR, –COOH, $–SO_3H$	deactivating	meta

- Strongly deactivated rings, such as nitrobenzene, do not undergo Friedel–Crafts reactions. Nitrobenzene is often used as the solvent for them.
- Halogens deactivate by −I but direct ortho/para by +R.

> **Solved example 1.** How would you tell but-1-yne from but-2-yne?
> Add ammoniacal $AgNO_3$. But-1-yne, a terminal alkyne, gives a white precipitate; but-2-yne does not react.

> **Solved example 2.** Where does nitration of toluene go?
> $–CH_3$ is ortho/para directing and activating, so the products are o- and p-nitrotoluene.

Traps. Only ethyne gives an aldehyde on hydration. Only terminal alkynes are acidic. Halogens are deactivating yet ortho/para directing. Friedel–Crafts fails on nitrobenzene. Benzene has 12 σ bonds, not 6.